

FinSpark Hackathon 2026

Premium Solution Proposal

Problem Statement 2: AI-Driven Correlation of Cybersecurity Telemetry & Transactional Behaviour

1. Executive Summary

Build an AI-powered Cyber Fusion Platform that correlates cyber telemetry, user behaviour and banking transactions into a single intelligence layer. The goal is to detect fraud earlier, reduce false positives and assist analysts with explainable AI.

2. Current Banking Challenges

Traditional fraud systems analyse transactions in isolation. Cybersecurity tools analyse only logs. This separation creates blind spots, delayed detection and alert fatigue.

3. Vision

Create a Bank Cognitive Security Brain capable of understanding relationships among customers, employees, devices, IPs, merchants and transactions.

4. End-to-End Architecture

Data Sources → Streaming Layer → Feature Store → AI Correlation Engine → Knowledge Graph → Explainable AI → Risk Dashboard → Automated Response.

5. Modules

Data Collector, Feature Engineering, Correlation Engine, Behaviour Engine, Fraud Detection, Insider Detection, Explainable AI, Executive Dashboard, Incident Manager.

6. AI Models

Isolation Forest, XGBoost, Autoencoder, LSTM, Graph Analytics, SHAP, LLM-based reasoning.

7. Dashboard

Live Alerts, Risk Timeline, Fraud Heatmap, Attack Story, Graph View, Executive KPI, Analyst Workspace.

8. Business Benefits

Reduced fraud losses, fewer false positives, faster investigations, improved customer trust, proactive cyber defence, regulatory compliance.

9. Scalability

Microservice architecture, REST APIs, Kafka-ready ingestion, cloud deployment, branch-wise scaling and modular AI services.

10. Future Roadmap

Predictive fraud prevention, quantum-safe monitoring, federated learning, autonomous SOC assistant and continuous model learning.

Top 10 Stand-Out Innovations

1. Digital DNA

Every customer gets a behavioural fingerprint generated from time, amount, merchant, device, location and transaction rhythm.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

2. Behaviour Twin

Maintain an AI twin and compare live activity with expected behaviour.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

3. Attack Story Generator

Automatically reconstruct the attack chain as a timeline instead of isolated alerts.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

4. Knowledge Graph

Connect customers, employees, devices, IPs and merchants visually.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

5. Predictive Threat Engine

Predict account takeover before fraudulent transfer occurs.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

6. Multi-Agent AI

Separate AI agents for Cyber, Fraud, Behaviour, Risk and Reporting.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

7. Trust Score

Dynamic trust score for user, device, merchant and IP.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

8. Quantum Readiness

Detect legacy encryption and estimate harvest-now-decrypt-later exposure.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

9. Executive Copilot

LLM answers natural-language questions from management.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

10. Adaptive Banking Immune System

Learns continuously from analyst feedback to reduce false positives.

Why it stands out: This feature directly improves business value, analyst productivity and cyber resilience while remaining practical for a banking deployment.

Suggested Prototype Flow

- 1 User logs in from a new device.

- 2 Cyber telemetry detects unfamiliar IP and device fingerprint.
- 3 Transaction engine observes unusually high transfer.
- 4 Knowledge Graph links the IP with earlier fraud cases.
- 5 AI Correlation Engine raises risk score from 18% to 96%.
- 6 LLM generates a human-readable explanation.
- 7 System recommends step-up authentication or transaction freeze.
- 8 Analyst reviews the attack story and approves action.

Technology Stack

Frontend: React/Streamlit
Backend: FastAPI
Database: PostgreSQL + Redis
AI: Scikit-learn, PyTorch, XGBoost, LLM (Gemini/OpenAI), NetworkX, SHAP.

Evaluation Mapping

Criteria	Weight	How PS2 Scores
Business Relevance	40%	Direct banking security platform with deployable value
Security	30%	Cyber + fraud + insider + explainable AI
Innovation	15%	Digital DNA, Multi-Agent AI, Knowledge Graph, Behaviour Twin
UX	5%	Interactive dashboard and AI Copilot
Scalability	5%	Microservice and cloud-ready
Maintainability	5%	Modular architecture

Problem Statement Comparison

Point	PS1	PS2
Business Impact	High	Very High
Innovation Potential	Medium	Excellent
Prototype Appeal	Good	Excellent
AI Scope	Medium	Very High
Future Ready	Medium	Excellent
Winning Potential	8/10	9.8/10

Final Recommendation

Problem Statement 2 provides the highest opportunity to impress the Bank of Maharashtra evaluation committee because it combines business value, cybersecurity impact, AI innovation and practical deployment. A polished prototype with Digital DNA, Behaviour Twin, Knowledge Graph and Multi-Agent AI would clearly differentiate the solution from conventional fraud-detection dashboards.